Chemistry 3A – Spring 2026 RETAKE VERSION

Sections 19320-19321

Midterm Examination #3

Indicate **BY DIRECTLY CIRCLING** CLEARLY on the exam page *one* and *only one* choice that is **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

1. Which law states that **pressure** and **volume** are inversely related, that is $P_1V_1 = P_2V_2$?
   (a) Boyle's Law (b) Charles' Law (c) Gay-Lussac's Law (d) Avogadro's Law
2. Convert a pressure of 1.50 atm to millimeters of mercury (mmHg)
   (a) 507 mmHg (b) 760 mmHg (c) 1,140 mmHg (d) 1,500 mmHg
3. In a redox reaction, the species that ***loses electrons*** is said to be:
   (a) reduced (b) precipitated (c) oxidized (d) unchanged
4. A chemical reaction where two or more simple reactants combine to form a single, more complex product (e.g., A + B → AB) is what type of reaction?
   (a) decomposition (b) single displacement (SD)
   (c) double displacement (DD) (d) combination (synthesis) reaction
5. What is molarity of aqueous solution in which 0.50 moles KCl dissolved to final 500.0 mL volume?
   (a) 0.25 M (b) 0.50 M (c) 1.0 M (d) 2.0 M
6. If you take 10.0 mL of a 2.5 M stock solution of NaCl and dilute it with water to a final volume of 50.0 mL, what is the final concentration (C_2)?
   (a) 0.10 M (b) 0.50 M (c) 1.25 M (d) 5.0 M
7. A solution is prepared by mixing 25.0 g of solute with 75.0 g of solvent. What is the percent by mass (m/m%) of the solute?
   (a) 25.0% (b) 33.3% (c) 75.0% (d) 100.0%
8. Calculate the **pressure** of n = 2.00 mol of gas at T = 273 K, V = 44.8 L. (R = 0.0821 L·atm/[mol·K])
   (a) 1.00 atm (b) 2.00 atm (c) 0.500 atm (d) 22.4 atm
9. Convert a **pressure** of 760 mmHg to atmospheres (atm).
   (a) 0.500 atm (b) 1.00 atm (c) 7.60 atm (d) 76.0 atm
10. Which of the following is a ***combustion*** reaction?
    (a) $CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$ (b) $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$
    (c) $HCl + NaOH \rightarrow NaCl + H_2O$ (d) $CaCO_3 \rightarrow CaO + CO_2$
11. Which of the following is the correct oxidation number of ***hydrogen*** in H_2O molecule?
    (a) 0 (b) -1 (c) +1 (d) +2
12. A substance that completely dissociates into ions when dissolved in an aqueous solution and conducts electricity strongly is classified as a:
    (a) non-electrolyte (b) weak electrolyte (c) strong electrolyte (d) saturated solution
13. What is the oxidation number (ON) for a neutral atom of Cl in its elemental state, Cl_2?
    (a) -1 (b) 0 (c) +1 (d) +7

14. The strongest type of ***intermolecular force*** that exists between two molecules of H_2O (water) is:
    (a) dispersion forces (London) (b) dipole-dipole forces
    (c) ion-dipole forces (d) hydrogen bonding

15. How many grams of KOH are required to prepare 1.00 L of a 0.500 M solution?
    (molar mass of KOH = 56.11 g/mol)
    (a) 5.61 g (b) 28.1 g (c) 56.1 g (d) 112.2 g

16. Which is a TRUE statement of the **Kinetic Molecular Theory** for an *ideal gas*?
    (a) gas molecules exert significant attractive forces on each other
    (b) the molecules that make up the gas phase must interact with their liquid phase
    (c) gas collisions are completely elastic (energy is not lost in collision)
    (d) the pressure of the gas is inversely proportional to temperature

17. Which of the following compounds is ***insoluble*** (forms precipitate) in water?
    (a) KNO_3 (b) Na_2SO_4 (c) AgCl (d) LiCl

18. A solution that has not yet reached its upper limit of dissolved solute is described as:
    (a) unsaturated (b) supersaturated (c) saturated (d) concentrated

19. A gas has initial pressure of 2.00 atm and volume of 5.00 L. If the volume is compressed to 2.50 L at constant temperature, what is the new pressure?
    (a) 1.00 atm (b) 4.00 atm (c) 5.00 atm (d) 10.0 atm

20. Hydrogen bonding can only occur between molecules where hydrogen is directly bonded to one of three highly electronegative atoms. These atoms are:
    (a) S, Cl, Br (b) Cl, Br, I (c) N, O, F (d) Na, K, Li

21. In a sugar-water solution, the sugar is the ***solute*** and the water is the ***solvent***. The <u>solvent</u> is defined as the component present in:
    (a) the smallest amount (b) the solid phase
    (c) the greatest amount (d) the ionic form

22. The concentration unit of ***molarity*** (***M***) is correctly expressed in which of the following units?
    (a) mol/kg (b) mol/L (c) gl/L (d) kg/mol

23. Which of the following is a spectator ion in the reaction (hint: compare complete ionic to net ionic):
    $NaCl\ (aq) + AgNO_3\ (aq) \rightarrow AgCl\ (s) + NaNO_3\ (aq)$?
    (a) Na^+ (b) Ag^+ (c) Cl^- (d) AgCl

24. According to statement best describes “like dissolves like”?
    (a) polar solutes dissolve in polar solvents (b) nonpolar solutes dissolve in nonpolar solvents
    (c) ionic solutes dissolve in polar solvents (d) all of choices (a) to (c)

25. A precipitate forms and you have to write the chemical reaction equation. How will you indicate the compound that precipitated?
    (a) **(*s*)** (b) **(*l*)** (c) **(*g*)** (d) **(*aq*)**

Circle either TRUE or FALSE whether the statement is true or false.

26. A solution that conducts electricity must contain ions.
True False

27. The reaction HCl (*aq*) + NaOH (*aq*) → H_2O (*l*) + NaCl (*aq*) is an acid-base neutralization reaction type.
True False

28. The balancing of a chemical reaction equation involves changing the subscripts of a chemical compound formula, like changing **$CuCl_2$** to **CuCl**
True False

29. The oxidation number of a pure element is always zero.
True False

30. In a chemical reaction equation, compounds to the ***right*** of the **reaction arrow** (yield symbon) are called ***reactants***.
True False

31. The oxidation number of any Group 1 metal (like Na or K) in any compound is always +1.
True False

32. The total pressure of a gas mixture is equal to the sum of the partial pressures of all the individual gases in the mixture.
True False

33. The solid that forms from a solution during a chemical reaction, often by mixing two solutions, is called a **precipitate**.
True False

34. At STP, the temperature is 0°C and pressure is 1 atm.
True False

35. The symbol ***n*** in the Ideal Gas Law equation (*PV=nRT*) is known as the ***universal gas constant***.
True False